Purpose:

The PCB is designed to control 6 high-power Bridgelux V22 LEDs in order to produce a light output that could illuminate a quarter acre of land with at least 5 foot-candles.

Each LED will be driven independently at a target current of 1.4A, using a MAX20078 synchronous buck LED driver circuit.

Power Distribution:

Power will enter the PCB through the main power connector at the top of the board and will pass through a fuse rated for 10A.

The PCB uses a four-layer design:

Layer 1- Top: This is where most of the components and electrical routing is.

Layer 2- GND: This is a grounding plane over the entire PCB.

Layer 3- +48V: This is a central fused power-distribution bus supplying the 6 drivers.

Layer 4- Bottom: This is for the additional electrical routing and where the LEDs will be soldered onto the board.

Vias will connect the outer-layer traces to the internal power (+48V) and ground (GND).

The estimated total input current is around 6.6A at 48V, assuming 33.9V for each LED (forward voltage per LED) and 90% conversion efficiency for the LED driver circuit.

LED Circuit Operation:

Each LED is powered by an independent MAX20078 synchronous buck converter circuit, which steps down the 48V input to the voltage required by the LED.

The controller operates two external MOSFETs (Q101 and Q102 in the first circuit) to regulate power delivered through the inductor (L101) to the LED.

Current Regulation: The CSP and CSN pins measure the voltage across the current-sense resistor (R105), located in the low-side MOSFET current path. The MAX20078 uses this measurement to regulate the average inductor current, therefore also the LED current.

Current Reference: The REFI pin controls the LED current setpoint. A reference-voltage divider is included only in the first circuit as we are keeping all LEDs at the same current, so we do not require more than one. The values for the resistors in the REFI reference-voltage divider are found by using an equation given by the data sheet of the chip using the current wanted and then plugging that number into a simple voltage divider equation.

Voltage Feedback: The R102/R103 divider provides a scaled output-voltage signal to the OUT pin, which is used by the controller.

Additional Control: The DIM pin provides PWM dimming control, while the bootstrap network supports high-side MOSFET switching. Input and output capacitors provide filtering and voltage stabilization.

All six of the circuits follow the same layout and design with them being grouped (101 - 601). The three circuits on the right are an almost mirror of the three on the left however due to the chip not being mirrorable they are slightly different from each other architecturally.

Physical PCB Layout:

The board as of right now measures at 85 X 170 mm with a target thickness of 1.6mm and four nominal 1 oz copper layers. The size is subject to change depending on screw holes being added however that would only change the size by about 5mm or so and would have no electrical changes.

The 6 LEDs are arranged in two columns of 3. Each LED has a 24 mm diameter cutout through the entire PCB as we plan to mount the LEDs on the back and have the light shine through the hole. The LEDs also come with two screw holes that will help to keep the LEDs in place.

The driver circuits are positioned as close to their respective LEDs to minimize routing distance and keep the high powered traces as short as possible. The main +48V power bus runs down the center of the PCB, with branches supplying the individual circuits.

Additional mounting holes and copper keepouts are included for mechanical attachment.

Review Notes:

Please feel free to tear this apart as this is my first time making a PCB and so I have been learning as I go. I know there are more than likely mistakes that I have overlooked and your help is greatly appreciated.